Two Giant Black Holes May Be Heading for a Crash

From: ScienceDaily Top Technology Mined: 2026-10-19 3 min read

Most big **galaxies** have a huge black hole in the middle. These **supermassive black holes** can weigh millions or even billions of times as much as the Sun. Scientists want to know how they grow so big. One idea is that two black holes can join together.

A team led by Silke Britzen in Germany studied a galaxy called Mrk 501. Its black hole shoots out a jet of tiny **particles** at nearly the speed of light. The team looked at **radio** data from about 23 years. They found a second jet moving in a different direction. That is a strong clue that a second black hole is there.

"It came as a complete surprise," Britzen said.

The team thinks the two black holes circle each other about every 121 days. They might crash together and **merge** in as little as 100 years. That is very fast for space!

Word Treasure

galaxies -- Galaxies are huge groups of stars, gas and dust held together by gravity.

supermassive black holes -- Supermassive black holes are giant black holes found in the middle of big galaxies.

particles -- Particles are extremely tiny bits of matter.

merge -- To merge means to join together into one.

radio -- Radio waves are a kind of invisible light that special telescopes can detect from space.

Background you need

A black hole is a place in space where gravity pulls so hard that not even light can get out.

Big galaxies, like our Milky Way, have a giant black hole in the middle.

When two galaxies crash together, their black holes may slowly move closer and join.

Break it down

WHO: A science team led by Silke Britzen in Germany
WHAT: Found clues that two supermassive black holes are circling each other
WHERE: In the middle of a galaxy called Mrk 501
WHY: Scientists want to learn how black holes grow so big, and joining together may be one way

Why it matters

It helps us understand the giant objects at the centers of galaxies, including our own.

Test what you remember

1. Where are supermassive black holes found?

- (A) On the Moon
- (B) Inside the Sun
- (C) In Earth's oceans
- (D) In the middle of most big galaxies

2. What is the name of the galaxy the team studied?

- (A) Kikai
- (B) Andromeda
- (C) Mrk 501
- (D) Milky Way

3. What clue showed there might be a second black hole?

- (A) A second jet moving in a different direction
- (B) A bright rainbow
- (C) A new moon
- (D) A loud sound

4. How many years of radio data did the team look at?

- (A) About 23
- (B) About 2
- (C) About 121
- (D) About 500

5. How often do the two black holes circle each other?

- (A) About every 121 days
- (B) Every 100 years
- (C) Every hour
- (D) Once a week

6. How soon might the black holes merge?

- (A) Never
- (B) Tomorrow
- (C) In as little as 100 years
- (D) In a billion years

Answer key (try the quiz first!): 1: D 2: C 3: A 4: A 5: A 6: C

Questions to think about

MY THOUGHTS (at least 20 words):